Shubham M

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Professional Summary

Computer Science and Engineering student at Bangalore Institute of Technology with a **9.06 CGPA**, skilled in Java, Data Structures and Algorithms, backend development, REST APIs, and databases. Hands-on experience building AI applications using Python, FastAPI, and React, with additional experience in Linux, Raspberry Pi, and IoT systems.

Technical Skills

Languages: Java, Python, JavaScript, HTML, CSS
Frameworks & Libraries: React, Node.js, FastAPI, LangChain, OpenCV
Databases: MySQL, Oracle SQL, MongoDB, SQLite, ChromaDB
AI/ML: Scikit-learn, XGBoost, Pandas, NumPy
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, TCP/IP
Tools & Platforms: Git, GitHub, Docker, Linux, Raspberry Pi, SSH, VNC

Projects

Enterprise RAG Assistant [GitHub] | *Python, FastAPI, React, LangChain, ChromaDB, Gemini*

- Developed a Retrieval-Augmented Generation (RAG) application for question answering over uploaded PDF documents using retrieved document context.
- Implemented document ingestion with PDF parsing, text chunking, embeddings, and semantic similarity search using ChromaDB and LangChain, integrated with Google Gemini.
- Built a React frontend and FastAPI backend supporting document uploads and conversational queries; configured Docker-based containerization for the application.

AI-Driven Post-Harvest Loss Prediction [GitHub] | *Python, FastAPI, Scikit-learn, XGBoost, Streamlit*

- Developed a machine learning application to predict crop spoilage risk and support storage-related decisions using agricultural data.
- Performed data preprocessing, feature engineering, and model training using Random Forest, XGBoost, and KNN algorithms.
- Integrated prediction models with FastAPI REST endpoints for crop-loss prediction, warehouse recommendations, and shelf-life estimation.

Internship Experience

Android & Raspberry Pi Development Intern — Sep 2025 – Nov 2025
InnovationHub — *Bangalore, India*

- Configured a Raspberry Pi with Linux for IoT-based edge computing and established remote access using SSH and VNC.
- Implemented TCP/IP socket communication to integrate Android applications with sensors and enable device-to-application data exchange.
- Developed an OpenCV-based image processing pipeline on embedded hardware to support real-time monitoring and automated alerts.

Education

Bangalore Institute of Technology (VTU) — Bangalore, India
B.E. in Computer Science and Engineering | ***CGPA: 9.06/10*** — *Expected 2027*
Jnanasudha PU College — Karkala, Karnataka
Pre-University Education (Science) | ***95%***

Achievements & Certifications

- **NPTEL Elite Certification:** Data Structures and Algorithms Using Java, IIT Kharagpur (2025).
- Solved **250+ LeetCode problems** in Data Structures and Algorithms. [LeetCode Profile]
- Participated in **IMPETUS 26.0**, organized by IEEE UVCE, in the Prompt & Pursuit event (2026).